A

PS-CVD5, main configuration: relative reduction in mean |SMD| vs PS-CVD5 base, %

	All 54 items	Vitals & core labs (9)	Other labs (10)	Echo: LV structure (7)	Echo: LV function (4)	Echo: diastolic/LA (8)	Echo: RV/ pulmonary (7)	Echo: valves/aorta (9)
+ ECG	**8.8***	**8.7***	6.9	14.9°	16.4°	**16.1***	−1.8	4.2
+ CLMBR-T	**12.4***	**22.7***	**24.1***	14.7°	1.0	0.7	0.2	**15.9***
+ ECG + CLMBR-T	**13.5***	**32.1***	**21.7***	**21.7***	7.6	10.4	−1.3	3.1
CLMBR-T vs ECG (2-sided)	3.9	15.3°	**18.4***	−0.2	−18.4°	−18.4°	1.9	12.2°
+ permuted ECG	1.0	2.2	1.7	−2.4	9.2	1.3	−5.0	0.6
+ permuted CLMBR-T	2.4	2.3	1.5	−2.0	7.0	1.4	2.8	3.7

B

PS-CVD5: CLMBR-T vs ECG, by configuration (positive = CLMBR-T better balanced)

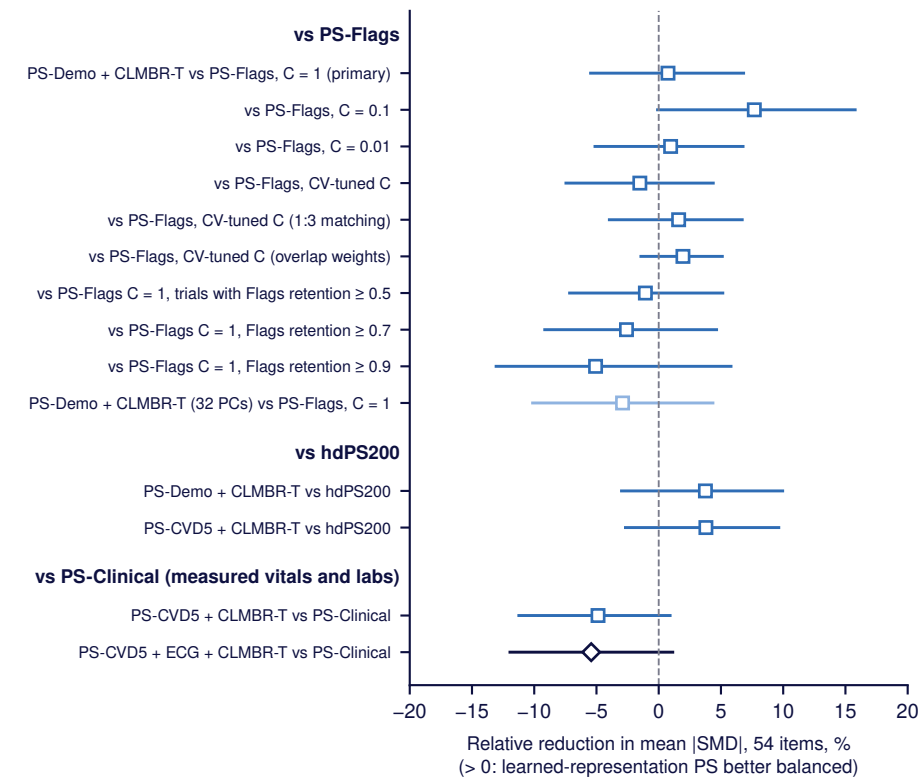
1:1, caliper 0.2 (main)	3.9	15.3°	**18.4***	−0.2	−18.4°	−18.4°	1.9	12.2°
1:1, caliper 0.1	2.5	**15.3***	16.8°	−0.9	−23.6°	−14.7	3.4	7.4
1:3 matching	**7.4***	**19.1***	**21.9***	−5.9	−4.6	−6.1	0.4	**12.9***
IPTW	−7.2	2.0	3.2	−16.0	−10.9	−29.7°	−8.1	−0.8
Overlap weights	**6.7***	**20.7***	**19.1***	−5.8	−4.6	−4.6	3.0	8.3°
Gradient-boosted PS	**8.1***	**19.4***	12.1	4.6	−11.6	10.1	3.6	9.2

* q < .05 (BH; overall column: balance family; domains: exploratory domain family)
° nominal P < .05 only. Domain differences are nominal; see legend.

−30 −20 −10 0 10 20 30
relative reduction, % (clipped at ±30)

C

Head-to-heads (S32, main unless stated; two-sided)



	Estimate (95% CI)	Better	P	q	|Δ log HR| ref → learned	Perm. P
vs PS-Flags						
PS-Demo + CLMBR-T vs PS-Flags, C = 1 (primary)	0.7 (−5.5 to 6.8)	13/31	.81	.86	0.186 → 0.174	.97
vs PS-Flags, C = 0.1	7.7 (−0.1 to 15.8)	20/32	.085	sens.	0.218 → 0.170	.32
vs PS-Flags, C = 0.01	1.0 (−5.1 to 6.8)	14/32	.76	sens.	0.202 → 0.170	.14
vs PS-Flags, CV-tuned C	−1.5 (−7.5 to 4.4)	13/32	.63	sens.	0.216 → 0.170	.001
vs PS-Flags, CV-tuned C (1:3 matching)	1.6 (−4.0 to 6.7)	18/32	.60	sens.	0.223 → 0.166	<.001
vs PS-Flags, CV-tuned C (overlap weights)	1.9 (−1.4 to 5.1)	18/32	.27	sens.	0.184 → 0.176	.70
vs PS-Flags C = 1, trials with Flags retention ≥ 0.5	−1.1 (−7.2 to 5.1)	10/27	.74	sens.	0.206 → 0.182	.82
vs PS-Flags C = 1, Flags retention ≥ 0.7	−2.6 (−9.2 to 4.6)	6/19	.50	sens.	0.222 → 0.211	.89
vs PS-Flags C = 1, Flags retention ≥ 0.9	−5.1 (−13.1 to 5.8)	3/11	.37	sens.	0.305 → 0.280	.77
PS-Demo + CLMBR-T (32 PCs) vs PS-Flags, C = 1	−2.9 (−10.1 to 4.4)	13/31	.44	.60	0.186 → 0.181	1.00
vs hdPS200						
PS-Demo + CLMBR-T vs hdPS200	3.8 (−3.0 to 10.0)	17/32	.29	.47	0.181 → 0.170	.095
PS-CVD5 + CLMBR-T vs hdPS200	3.8 (−2.7 to 9.6)	19/32	.26	.43	0.181 → 0.166	.15
vs PS-Clinical (measured vitals and labs)						
PS-CVD5 + CLMBR-T vs PS-Clinical	−4.9 (−11.2 to 0.9)	9/32	.12	.25	0.158 → 0.166	.19
PS-CVD5 + ECG + CLMBR-T vs PS-Clinical	−5.4 (−11.9 to 1.1)	13/32	.13	.27	0.158 → 0.158	.17